

A Practical Guide to Perceptron

TOPIC -- PERCEPTRON

$$y = \text{sign}(w \cdot x + b)$$

$$w \leftarrow w + \eta(y - \hat{y})x$$

-- 01 --

While the perceptron algorithm is guaranteed to converge on some solution in the case of a linearly separable training set, it may still pick any solution and problems may admit many solutions of varying quality. The perceptron of optimal stability, nowadays better known as the linear support-vector machine, was designed to solve this problem (Krauth and Mezard, 1987).

-- 02 --

Perceptrons (Minsky and Papert, 1969) studied the kind of perceptron networks necessary to learn various Boolean functions. Consider a perceptron network with n input units, one hidden layer, and one output, similar to the Mark I Perceptron machine. It computes a Boolean function of type $f : 2^n \rightarrow 2$. They call a function conjunctively local of order k , iff there exists a perceptron network such that each unit in the hidden layer connects to at most k input units. Theorem. (Theorem 3.1.1): The parity function is conjunctively local of order n . Theorem. (Section 5.5): The connectedness function is conjunctively local of order $\Omega(n^{1/2})$.

-- 03 --

A Perceptron implemented in MATLAB to learn binary NAND function Archived 2012-11-06 at the Wayback Machine Chapter 3 Weighted networks - the perceptron and chapter 4 Perceptron learning of Neural Networks - A Systematic Introduction by Raúl Rojas (ISBN 978-3-540-60505-8) History of perceptrons Mathematics of multilayer perceptrons Applying a perceptron model using scikit-learn - https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Perceptron.html

-- 04 --

In a 1958 press conference organized by the US Navy, Rosenblatt made statements about the perceptron that caused a heated controversy among the fledgling AI community; based on Rosenblatt's statements, The New York Times reported the perceptron to be "the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence." The Photo Division of Central Intelligence Agency, from 1960 to 1964, studied the use of Mark I Perceptron machine for recognizing militarily interesting silhouetted targets (such as planes and ships) in aerial photos.

-- 05 --

Learning a Boolean function Consider a dataset where the x are from $\{-1, +1\}^n$, that is, the vertices of an n -dimensional hypercube centered at origin, and $y = \theta(x_i)$. That is, all data points with positive x_i have $y = 1$, and vice versa. By the perceptron convergence theorem, a perceptron would converge after making at most n mistakes. If we were to write a logical program to perform the same task, each positive example shows that one of the coordinates is the right one, and each negative example shows that its complement is a positive example. By collecting all the known positive examples, we eventually eliminate all but one coordinate, at which point the dataset is learned. This bound is asymptotically tight in terms of the worst-case. In the worst-case, the first presented example is entirely new, and gives n bits of information, but each subsequent example would differ minimally from previous examples, and gives 1 bit each. After $n + 1$ examples, there are $2n$ bits of information, which is sufficient for the perceptron (with $2n$ bits of information). However, it is not tight in terms of expectation if the examples are presented uniformly at random, since the first would give n bits, the second $n/2$ bits, and so on, taking $O(\ln n)$ examples in total.

-- 06 --

In machine learning, the perceptron is an algorithm for supervised learning of binary classifiers. A binary classifier is a function that can decide whether or not an input, represented by a vector of numbers, belongs to some specific class. It is a type of linear classifier, i.e. a classification algorithm that makes its predictions based on a linear predictor function combining a set of weights with the feature vector.

-- 07 --

η is the learning rate of the perceptron. Learning rate is a positive number usually chosen to be less than 1. The larger the value, the greater the chance for volatility in the weight changes.

-- 08 --

A single perceptron can learn to classify any half-space. It cannot solve any linearly nonseparable vectors, such as the Boolean exclusive-or problem (the famous "XOR problem"). A perceptron network with one hidden layer can learn to classify any compact subset arbitrarily closely. Similarly, it can also approximate any compactly-supported continuous function arbitrarily closely. This is essentially a special case of the theorems by George Cybenko and Kurt Hornik.

-- 09 --

Variants The pocket algorithm with ratchet (Gallant, 1990) solves the stability problem of perceptron learning by keeping the best solution seen so far "in its pocket". The pocket algorithm then returns the solution in the pocket, rather than the last solution. It can be used also for non-separable data sets, where the aim is to find a perceptron with a small number of misclassifications. However, these solutions appear purely stochastically and hence the pocket algorithm neither approaches them gradually in the course of learning, nor are they guaranteed to show up within a given number of learning steps. The Maxover algorithm (Wendemuth, 1995) is "robust" in the sense that it will converge regardless of (prior) knowledge of linear separability of the data set. In the linearly separable case, it will solve the training problem – if desired, even with optimal stability (maximum margin between the classes). For non-separable data sets, it will return a solution with a computable small number of misclassifications. In all cases, the algorithm gradually approaches the solution in the course of learning, without memorizing previous states and without stochastic jumps. Convergence is to global optimality for separable data sets and to local optimality for non-separable data sets. The Voted Perceptron (Freund and Schapire, 1999), is a variant using multiple weighted perceptrons. The algorithm starts a new perceptron every time an example is wrongly classified, initializing the weights vector with the final weights of the last perceptron. Each perceptron will also be given another weight corresponding to how many examples do they correctly classify before wrongly classifying one, and at the end the output will be a weighted vote on all perceptrons. In separable problems, perceptron training can also aim at finding the largest separating margin between the classes. The so-called perceptron of optimal stability can be determined by means of iterative

training and optimization schemes, such as the Min-Over algorithm (Krauth and Mezard, 1987) or the AdaTron (Anlauf and Biehl, 1989)). AdaTron uses the fact that the corresponding quadratic optimization problem is convex. The perceptron of optimal stability, together with the kernel trick, are the conceptual foundations of the support-vector machine. The α -perceptron further used a pre-processing layer of fixed random weights, with thresholded output units. This enabled the perceptron to classify analogue patterns, by projecting them into a binary space. In fact, for a projection space of sufficiently high dimension, patterns can become linearly separable. Another way to solve nonlinear problems without using multiple layers is to use higher-order networks (sigma-pi unit). In this type of network, each element in the input vector is extended with each pairwise combination of multiplied inputs (second order). This can be extended to an n-order network. A generalization of the perceptron model is the Receptron that incorporates non-linear interactions between inputs. A single receptron is able to classify non-linear Boolean functions. It should be kept in mind, however, that the best classifier is not necessarily that which classifies all the training data perfectly. Indeed, if we had the prior constraint that the data come from equi-variant Gaussian distributions, the linear separation in the input space is optimal, and the nonlinear solution is overfitted. Other linear classification algorithms include Winnow, support-vector machine, and logistic regression.